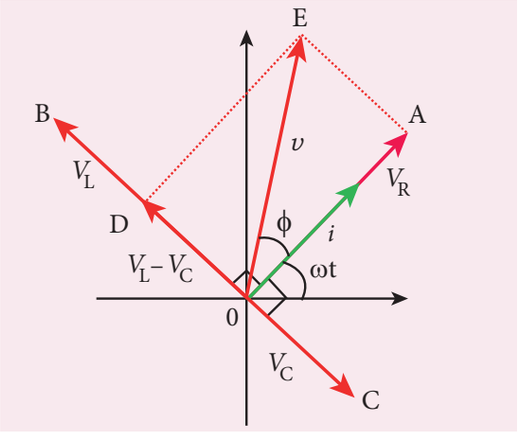


Figure 4.47 Phasor diagram for a series *RLC* – circuit when $V_L > V_C$

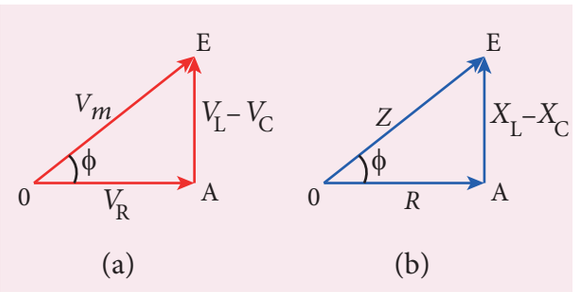


Figure 4.48 Voltage and impedance triangle when $X_L > X_C$

The voltage triangle and impedance triangle are given in the Figure 4.48.

From phasor diagram, the phase angle between v and i is found out from the following relation

$$\tan\phi = \frac{V_L - V_C}{V_R} = \frac{X_L - X_C}{R} \qquad (4.48)$$

Special cases

(i) If $X_L > X_C$, $(X_L - X_C)$ is positive and phase angle ϕ is also positive. It means that the applied voltage leads the current by ϕ (or current lags behind voltage by ϕ). The circuit is inductive.

$$\therefore v = V_m \sin\omega t;\ i = I_m \sin(\omega t - \phi)$$

(ii) If $X_L < X_C$, $(X_L - X_C)$ is negative and ϕ is also negative. Therefore current leads voltage by ϕ and the circuit is capacitive.

$$\therefore v = V_m \sin\omega t;\ i = I_m \sin(\omega t + \phi)$$

(iii) If $X_L = X_C$, ϕ is zero. Therefore current and voltage are in the same phase and the circuit is resistive.

$$\therefore v = V_m \sin\omega t;\quad i = I_m \sin\omega t$$

4.7.7 Resonance in series RLC Circuit

When the frequency of the applied alternating source (ω_r) is equal to the natural frequency $\left[1/\sqrt{LC}\right]$ of the *RLC* circuit, the current in the circuit reaches its maximum value. Then the circuit is said to be in **electrical resonance**. The frequency at which resonance takes place is called **resonant frequency**.

Table 4.1 Summary of results of AC circuits

Type of Impedance	Value of Impedance	Phase angle of current with voltage	Power factor
Resistance	R	0°	1
Inductance	$X_L = \omega L$	90° lag	0
Capacitance	$X_C = 1/\omega C$	90° lead	0
R- L - C	$\sqrt{R^2 + \left(\omega L - 1/\omega C\right)^2}$	Between 0° and 90° lag or lead	Between 0 and 1

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